

Medication & Vaccination Tracking System Implementation

Overview

The system now supports comprehensive medication and vaccination tracking for broiler production cycles, meeting all specified requirements.

Requirements Implemented

1. Record medication name, dosage, date, administered by

- **Models:** `MedicationRecord` interface with fields:
 - `medicationName`, `dosage`, `date`, `administeredBy`
 - Additional fields: `medicationType`, `administrationRoute`, `birdsCount`, `cost`, `complianceCheck`
- **API Endpoints:**
 - `POST /api/medication/records` - Create medication record
 - `GET /api/medication/records?cycleId={id}` - Get records for cycle

2. Track vaccination schedules and compliance

- **Models:** `MedicationSchedule` interface with:
 - `scheduleType` (single/recurring), `startDate`, `endDate`, `frequencyDays`, `targetDay`
 - `administered`, `administeredDate`, `administeredBy` for tracking completion
- **Compliance Tracking:**
 - `calculateComplianceRate()` function to compute compliance percentage
 - `MedicationComplianceReport` interface for comprehensive reports
- **API Endpoints:**
 - `GET /api/medication/schedules?cycleId={id}` - Get schedules
 - `GET /api/medication/reports/compliance/{cycleId}` - Generate compliance report

3. Generate medication reports for regulatory compliance

- **Report Types:**
 - `MedicationComplianceReport`: Shows scheduled vs administered medications
 - `MedicationAnalytics`: Usage statistics, costs, and trends
 - Withdrawal period violation detection
- **API Endpoints:**
 - `GET /api/medication/reports/compliance/{cycleId}` - Compliance report
 - `GET /api/medication/analytics` - Analytics report

4. Alert for upcoming vaccination schedules

- **Alert System:**
 - `getUpcomingSchedules()`: Returns schedules due in next X days

- `getMedicationAlerts()`: Comprehensive alert system including:
 - Upcoming schedules (next 3 days)
 - Low stock medications
 - Expired medications
 - Compliance issues (<80% compliance rate)
- **API Endpoint:** `GET /api/medication/alerts`

5. Track medication inventory and usage

- **Inventory Management:**
 - `MedicationInventory` interface with:
 - `batchNumber`, `expiryDate`, `quantity`, `unitCost`, `currentStock`
 - `reorderLevel`, `lastRestocked`, `supplier`
 - Automatic stock deduction when medications are administered
 - Low stock and expired medication detection
- **API Endpoints:**
 - `GET /api/medication/inventory` - List all inventory
 - `GET /api/medication/inventory/low-stock` - Low stock items
 - `GET /api/medication/inventory/expired` - Expired items
 - `POST /api/medication/inventory` - Add new inventory
 - `PUT /api/medication/inventory/{id}/stock` - Update stock levels

Technical Implementation

Backend Structure

1. **Models** (`src/models/Medication.ts`):

- `MedicationType` enum: ANTIBIOTIC, VACCINE, VITAMIN, DEWORMER, etc.
- `AdministrationRoute` enum: ORAL, INJECTION, WATER, FEED, TOPICAL
- Interfaces: `MedicationSchedule`, `MedicationRecord`, `MedicationInventory`
- Report interfaces: `MedicationComplianceReport`, `MedicationAnalytics`

2. **Service** (`src/services/medicationService.ts`):

- `MedicationService` class with static methods
- CRUD operations for schedules, records, and inventory
- Analytics and reporting functions
- Alert generation system

3. **Routes** (`src/routes/medication.ts`):

- RESTful API endpoints with authentication
- Input validation and error handling
- Support for filtering and query parameters

4. **Integration** (`src/app.ts`):

- Added medication routes to main application

- Updated root endpoint documentation

Frontend Integration

1. **API Service** (`src/services/api.ts`):

- Added medication-related methods to `ApiService` class
- Methods for schedules, records, inventory, reports, analytics, and alerts
- Type-safe API calls with proper error handling

2. **Types**:

- Enums for medication types and administration routes
- Comprehensive TypeScript interfaces for all data structures

Key Features

1. Comprehensive Tracking

- Medication schedules with flexible recurrence patterns
- Detailed administration records with compliance checks
- Inventory management with batch tracking and expiry dates

2. Regulatory Compliance

- Withdrawal period monitoring to prevent violations
- Compliance rate calculations and reporting
- Audit trails for all medication activities

3. Proactive Alerts

- Upcoming schedule notifications
- Low stock warnings
- Expired medication alerts
- Compliance issue detection

4. Analytics & Reporting

- Usage statistics by medication type
- Cost analysis and trends
- Monthly usage reports
- Top medication tracking

5. Integration with Production Cycles

- Medication schedules linked to specific production cycles
- Automatic bird count tracking
- Integration with existing production planning system

Usage Examples

Creating a Medication Schedule

```
const schedule = await apiService.createMedicationSchedule({
  cycleId: 'cycle-123',
  medicationName: 'Newcastle Vaccine',
  medicationType: 'vaccine',
  dosage: '0.5ml/bird',
  administrationRoute: 'injection',
  scheduleType: 'single',
  startDate: '2026-01-25',
  birdsCount: 5000,
  lastUpdatedBy: 'user-123'
});
```

Recording Medication Administration

```
const record = await apiService.createMedicationRecord({
  scheduleId: 'schedule-456',
  cycleId: 'cycle-123',
  medicationName: 'Newcastle Vaccine',
  medicationType: 'vaccine',
  dosage: '0.5ml/bird',
  administrationRoute: 'injection',
  date: '2026-01-25',
  administeredBy: 'user-123',
  birdsCount: 5000,
  cost: 250.00,
  currency: 'USD',
  lastUpdatedBy: 'user-123'
});
```

Getting Compliance Report

```
const report = await apiService.getMedicationComplianceReport('cycle-123');
console.log(`Compliance Rate: ${report.complianceRate}%`);
console.log(`Scheduled: ${report.scheduledMedications}`);
console.log(`Administered: ${report.administeredMedications}`);
```

Checking Alerts

```
const alerts = await apiService.getMedicationAlerts();
console.log(`Upcoming schedules: ${alerts.upcomingSchedules.length}`);
console.log(`Low stock items: ${alerts.lowStock.length}`);
console.log(`Expired medications: ${alerts.expired.length}`);
```

Testing

A comprehensive test script (`test-medication.js`) is provided to verify all functionality:

- Medication schedule creation and retrieval
- Medication record creation and tracking
- Inventory management operations
- Alert generation
- Analytics and reporting

Files Created/Modified

New Files

1. `src/models/Medication.ts` - Data models and interfaces
2. `src/services/medicationService.ts` - Business logic service
3. `src/routes/medication.ts` - API routes
4. `test-medication.js` - Test script
5. `MEDICATION_TRACKING_IMPLEMENTATION.md` - This documentation

Modified Files

1. `src/app.ts` - Added medication routes
2. `src/services/api.ts` - Added medication API methods

Next Steps

1. **Frontend UI Components:** Create React components for medication management
2. **Dashboard Integration:** Add medication alerts to main dashboard
3. **Export Features:** PDF/Excel export for compliance reports
4. **Mobile Support:** Offline medication recording for field workers
5. **Integration Testing:** End-to-end testing with production data

Conclusion

The medication and vaccination tracking system has been successfully implemented with all required features. The system provides comprehensive tracking, regulatory compliance support, proactive alerts, and detailed reporting capabilities, fully meeting the specified requirements for broiler production cycle management.